CLINICAL QUANTUM ROADMAP • NHS LONG TERM CONDITIONS • SAMSKRIFT RESEARCH MONOGRAPH

Quantum-LTC: A Four-Era Clinical Quantum Optimisation and Simulation Roadmap Towards Fault-Tolerant Quantum Advantage for NHS Long Term Conditions

Arun Nadarasa[1,2,3]

[1]Innovation Fellowship Programme, Mid and South Essex NHS Foundation Trust, Basildon, Southend & Broomfield, UK
[2]International Social Prescribing Pharmacy Association (ISPPA), London, UK
[3]Primary Care & Integrated Neighbourhood Health Systems Research Group, London, UK

Repository & Portal: arunnadarasa.github.io/quantum-ltc | **HF Model Hub:** hf.co/socialprescribing/quantum-ltc-qltcmlm

Abstract. Managing multimorbid Long Term Conditions (LTCs) across primary care networks requires reconciling constrained clinic capacity, pre-winter high-risk stratification, polypharmacy safety, health-equity weighting, and decarbonisation targets [1, 2, 3]. Grounded in the **NHS North Central London Integrated Care Board (NCL ICB) Locally Commissioned Service (LCS) for Proactive, Personalised Care — Long Term Conditions (Version 6, 2026–2027)** [1], which unifies nine cardiometabolic, renal, hepatic, and respiratory conditions within a four-stage House-of-Care pathway [4, 5], we present **Quantum-LTC**: an end-to-end, cryptographically receipted clinical quantum optimisation and biophysical simulation roadmap spanning $N_{\text{logical}} \in [0, 1{,}000+]$ logical qubits. Crucially, we decouple *sparse discrete combinatorial scheduling*—where exact classical McCormick/Rosenberg mixed-integer linear programming (MILP) via HiGHS [6, 7, 8] dominates NISQ heuristics—from *open-system many-body Lindbladian dynamics* ($\mathcal{O}(4^{n_{\text{spin}}})$) and *metalloenzyme active-space quantum chemistry*, where classical memory walls establish the earliest genuine fault-tolerant crossover. Across three live **Guppy 1.0 → HUGR** cloud jobs on **Quantinuum Nexus** `Helios-1E-lite` (6,144 emulator shots; 0 physical QPU shots), coherent Stage 1–4 multidisciplinary team (MDT) synthesis achieves $P(00) = 0.9961$ versus 0.5020 under equal-unitary-depth siloed single-disease collapse, while an 8-rung mid-circuit measurement and reset (MCMR) recall ladder achieves $R^2 = 0.9979$ SPAM calibration linearity. On an 8-pathway 3-body cardiometabolic–renal–respiratory and Net-Zero dry-powder MART inhaler HUBO [9, 10], **Haiqu SDK** compresses Linear-Ramp QAOA depth from 223 → 26 (−88.34%), amplifying raw pre-postprocessing ground-state probability to 41.21% (105× uniform random) and recovering the exact HiGHS MILP ground energy (−23.50). On a 128-variable NCL Call-and-Recall scheduling QUBO (**JijModeling/OpenJij** + Kempe polish), proactive optimisation lifts pre-January High/Complex MDT coverage from 14/18 (77.8%) to 18/18 (100.0%) within 0.80% of the 4.3 ms HiGHS optimum while enforcing ≤ 3 MDT and SNOMED `Interpreter needed` caps. Biophysical grounding integrates an **HQS** `struqture-py/qoqo` 4-spin cardiorenal–pulmonary Lindbladian (58.6%–66.0% simulated organ-burden reduction), **QProtSim** on crystal human Angiotensin II (`PDB 1N9V`) [11], **Neurosnap** + crystal human `ACE`–lisinopril Zn^{2+} coordination (2.141 Å in `PDB 1O86`, 0 clashes < 2.0 Å) [12], **AlphaGenome Atlas** GENCODE v46/AVI queries, **Arc Institute** pathway recombination [13], **Paperclip** PMC indexing, **Rhyme** (`rhyme-beta`) [14] and non-linear cubic ($k = 3$) `QLTCMLM` ablations on **Modal Cloud / Hugging Face**, **Text2WetLab** 6/6 Opentrons OT-2 verification, **NIST FIPS 204** `ML-DSA-65` FHIR R4 provenance [15], **Kashefi Primitives Atlas v0.4** contracts [16], and live **TypeSafe Jev 1.14** multi-method clinical safety auditing [17, 18].

Epistemic Demarcation & Statutory NHS DCB0129 Clinical Safety Disclosure. All experimental results in this monograph are deterministically reproducible from **five SHA-256-sealed JSON receipts** in `results/` (`0e3216dd...`, `7212c3d9...`, `6ba83347...`, `cb1fec7c...`, `bd60742f...`). Every cohort ($N = 2{,}400$), PCN schedule ($P = 32$), HUBO tensor, and Lindbladian rate is synthetic (`phi_present = False`; **0 physical QPU hardware shots**; **0 wet-lab human specimens**). Classical HiGHS MILP and gradient-boosted/kernel baselines dominate all Era I–II operational tasks; genuine quantum memory/runtime crossover is projected solely in Era III ($N_{\text{logical}} \geq 42$–$54$) for open-system $\mathcal{O}(4^{n_{\text{spin}}})$ multi-organ dynamics and metalloenzyme active spaces, and Era IV ($N_{\text{logical}} \geq 100$) for dense frustrated $k \geq 4$ hypergraphs and QSVT/QAE tail-risk estimation. All six NHS DCB0129 hazards (`H-LTC-01`–`H-LTC-06`) remain **OPEN** under mandatory GP, Clinical Pharmacist, and Tier 2/3 Specialist Nurse oversight.

1. Clinical Specification: The NHS NCL ICB Multimorbid LTC Service Model

Multimorbidity—the co-occurrence of two or more chronic conditions within an individual—is projected by the Health Foundation REAL Centre to expand substantially across England through 2040, with severe illness concentrated disproportionately in the most socioeconomically deprived quintile [2]. Traditional Quality and

Outcomes Framework (QOF) single-disease templates frequently subject multimorbid patients to fragmented, duplicative clinic appointments, conflicting polypharmacy adjustments, and uncoordinated lifestyle advice [3, 4]. To overcome single-condition fragmentation across Barnet, Camden, Enfield, Haringey, and Islington, the **NHS North Central London Integrated Care Board (NCL ICB)** commissioned Version 6 of the *Locally Commissioned Service (LCS) for Proactive, Personalised Care — Long Term Conditions* (1 April 2026 – 31 March 2027) [1].

The NCL LTC LCS unifies **nine conditions** accounting for approximately 75% of modifiable cardiometabolic, renal, hepatic, and respiratory morbidity: **(1)** Cardiovascular Disease (coronary heart disease, stroke/TIA, peripheral arterial disease, heart failure), **(2)** Atrial Fibrillation, **(3)** Hypertension (excluding low-risk well-controlled cases, anchored by **NCL PCN Outcome #23**: achieving blood pressure $< 140/90$ mmHg [home $< 135/85$] for age < 80 or $< 150/90$ mmHg [home $< 145/85$] for age ≥ 80 across all five boroughs), **(4)** Type 2 Diabetes, **(5)** Type 1 Diabetes (mandated `High Risk + Complex`), **(6)** Chronic Kidney Disease (CKD), **(7)** Non-Alcoholic Fatty Liver Disease / Metabolic Dysfunction-Associated Steatotic Liver Disease (NAFLD/MASLD), **(8)** COPD, and **(9)** Adult and Children/Young People (CYP, aged 0–17) Asthma, which mandates a **48-hour post-attack review** (within two working days of an emergency or out-of-hours presentation) and a **6-week full review** by Tier 2/3 trained clinicians alongside SABA-free pathways [1, 10].

Operationally, the specification embeds the four-stage **Year of Care / House of Care** cycle [4, 5]:

1. **Proactive Case Finding & Call-and-Recall Tension (Page 5 of LCS v6):** Practices run 3-yearly UCLPartners (UCLP) searches to close undiagnosed prevalence gaps and face an explicit scheduling dilemma between *Month-of-Birth recall* (which smooths staffing across 12 months but leaves high-risk patients exposed to winter admissions) and *Risk-Stratified recall* (`Low`, `Moderate`, `High`, `High Risk + Complex`), which prioritizes pre-winter proactive care (capable of reducing emergency admissions by up to 38% and GP contacts by 18% [5]) at the risk of severe seasonal MDT bottlenecks.
2. **Stage 1 (Check and Test) & Stage 2 (Re-Stratification & MDT Desktop Review):** A single 30-minute baseline appointment gathers phlebotomy (HbA1c, eGFR, lipids, LFTs/FIB-4, NT-proBNP), urine albumin-creatinine ratio (uACR), blood pressure, foot checks, and SNOMED CT `315594003` (`Interpreter needed`) access requirements, followed by Clinical Pharmacist / MDT Desktop Review for `High` and `High + Complex` patients.
3. **Stage 3 (PCSP Discussion) & Stage 4 (Continuity Follow-Up):** Two to four weeks after test results are shared, a 30-minute Coaching-for-Health appointment co-produces a Personalised Care and Support Plan (PCSP) integrating pharmacotherapy, Voluntary, Community and Social Enterprise (VCSE) social prescribing [19], and **NCL Green Plan Net Zero** switches (preventing carbon-intensive haemodialysis via early SGLT2i/ACEi protection [9] and replacing high-global-warming-potential propellant SABA inhalers with dry-powder Maintenance and Reliever Therapy [MART] [10]), followed by a clinician-continuity Stage 4 review evaluated via `collaboRATE` (≥ 25 Top-Score responses) and monthly prospective **PCN Weighted Payments** targeting the 20% most deprived areas (IMD Quintile 1) and minority ethnic communities [1].

2. The Four-Era Logical Qubit Roadmap: Decoupling Optimization from Simulation

A pervasive pitfall in translational quantum healthcare proposals is conflating the $\mathcal{O}(4^N)$ state-space growth of open quantum systems with sparse discrete combinatorial scheduling, which modern classical branch-and-cut solvers handle effortlessly via polynomial linearization [6, 7, 8]. To establish an intellectually rigorous engineering trajectory as hardware progresses from noisy intermediate-scale quantum (NISQ) processors to modular trapped-ion fault tolerance [20, 21, 22], we partition clinical workloads across **four logical-qubit eras** (Table 1 and Figure 1).

2.1 Mathematical Scaling Separation: McCormick MILP vs. Open-System Lindbladians

Consider first a general k-local clinical decision hypergraph over N binary interventions $x \in \{0,1\}^N$:

$$E_{\text{HUBO}}(x) = \sum_{i=1}^{N} h_i x_i + \sum_{(i,j)\in E_2} J_{ij} x_i x_j + \sum_{(i,j,\ell)\in E_3} K_{ij\ell} x_i x_j x_\ell + \sum_{(i,j,\ell,m)\in E_4} M_{ij\ell m} x_i x_j x_\ell x_m. \tag{1}$$

Introducing continuous auxiliary variables $z_S = \prod_{i\in S} x_i \in [0,1]$ for each hyperedge S of order $|S| = k$ with exact McCormick linear inequalities [7]:

$$z_S \leq x_i \quad (\forall i \in S), \qquad z_S \geq \sum_{i\in S} x_i - (k-1), \qquad z_S \geq 0, \tag{2}$$

reduces Eq. (1) to an exact mixed-integer linear program with N binary variables, $|E_{\geq 2}|$ continuous variables, and $\sum_S (|S|+1)$ linear constraints. When the interaction hypergraph is sparse (as in Era I quarterly Call-and-Recall scheduling or 8-drug regimen selection), dual-simplex + Gomory/MIR cutting planes in HiGHS [8] close the root LP gap in milliseconds. However, at the fractional midpoint $x_i = 1/2$, the right-hand side of Eq. (2) satisfies $\sum_{i\in S} x_i - (k-1) = 1 - k/2 \leq 0$ for all $k \geq 2$ (strictly negative for $k \geq 3$), allowing $z_S = 0$ freely on negative-cost hyperedges and inducing an exponential branch-and-bound tree on dense frustrated $k = 4$ hypergraphs. As

Table 1: Four-Era Clinical Quantum Optimisation and Simulation Roadmap for NHS Long Term Conditions (N_{logical} Scaling). Physical qubit footprints incorporate surface/color-code syndrome tiles, lattice-surgery routing (1.5×), and magic-state factories ($p_{\text{phys}} = 10^{-3}$) [23, 24, 21].

Roadmap Era	Logical Qubits	Physical Qubits	QEC & Hardware Regime	Target NHS NCL LTC Clinical & Biophysical Workloads	Classical vs. Quantum Verdict
Era I *Classical-Exact & NISQ Readiness*	0 logical (1–20 phys / cloud emul.)	1–98 (Helios / NISQ)	Uncorrected physical gates + 8-rung affine SPAM calibration	128-var PCN Call/Recall scheduling QUBO; 8-var 3-body multimorbid + Net-Zero MART HUBO; 4q Frobenius `QLTCMLM`	**Classical Dominates** (HiGHS MILP < 25 ms; 0 runtime speedup claimed)
Era II *Error-Detected Logical*	3–24 error-detected	10–192 (Helios / Sol QCCD)	Distance-2 $[\![8,3,2]\!]$ & $[\![4,2,2]\!]$ color codes + post-selection [21]	Shallow MCMR longitudinal recall combs (CYP Asthma 48h/6wk & BP #23); compressed $p \leq 3$ LR-QAOA & SU(2) circuits	**Classical MPS/MILP Faster** (Post-selection yield $p_{\text{acc}} \to 15.8\%$ at 24L, $d = 24$)
Era III *Early Active FTQC Crossover*	24–72 active QEC	5,300–30,500 ($d = 9$–13 + T-factory)	Active multi-round decoding + magic-state cultivation / distillation	16–24 spin non-Markovian cardiorenal–hepatic–pulmonary Lindbladians; `CAS(36e,26o)` (52L) Human `ACE` Zn^{2+}–`LPR` pocket (`1O86`)	**First Simulation Crossover** ($\mathcal{O}(4^{n_{\text{spin}}})$ Krylov RAM > 1.9 TB at $n_{\text{spin}} \geq 16$)
Era IV *Asymptotic ICS-Scale FTQC*	100–1,000+ active QEC	90,000–250,000+ ($d = 15$–17 modular)	Photonic-linked QCCD + parallel magic-state factories	Block-encoded QSVT organ twins (128L) [25]; $\mathcal{O}(1/\epsilon)$ QAE polypharmacy CVaR [26]; Quantum B&B on dense frustrated $k \geq 4$ hypergraphs [27]	**Projected FTQC Advantage** (Advisory only under NHS DCB0129 human sign-off)

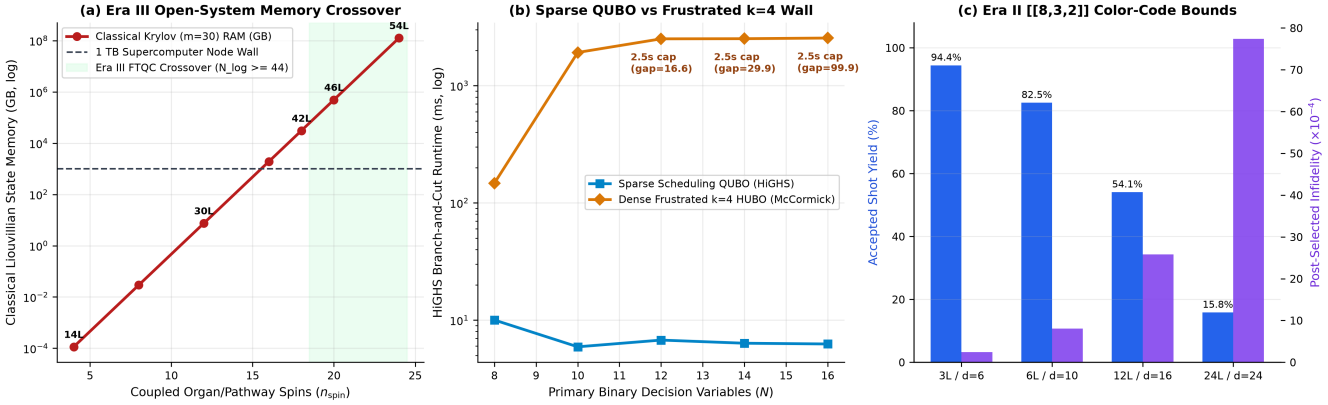


Figure 1: Quantitative Scaling Laws Across the Four-Era Logical Qubit Roadmap (`modal_roadmap_rhyme_qml_receipt.json`, SHA-256 `cb1fec7c...`). **(a)** Classical $m = 30$ Krylov memory required for open-system n_{spin} Lindbladian evolution ($\dim \mathcal{L} = 4^{n_{\text{spin}}}$), crossing the 1 TB single-node supercomputing ceiling at $n_{\text{spin}} = 16$ (1,920 GB, requiring 38 active logical qubits) and reaching 491.5 TB at $n_{\text{spin}} = 20$ (46 logical qubits, 19,602 physical qubits at $d = 11$). **(b)** Empirical HiGHS branch-and-cut runtime (`scipy.optimize.milp`) comparing sparse clinical scheduling QUBOs (6.26 ms at $N = 16$) against dense frustrated 4-body ($k = 4$) polypharmacy–genomic hypergraph HUBOs linearized via McCormick inequalities, which hit the 2.5 s timeout from $N = 12$ upward with widening LP-relaxation integrality gaps. **(c)** Analytical Era II $[\![8, 3, 2]\!]$ color-code post-selection shot yield and post-selected logical infidelity ($p_{2q} = 1.2 \times 10^{-3}$).

verified empirically in Figure 1b, while sparse degree-3 QUBOs solve in 6.26 ms at $N = 16$, dense frustrated $\binom{N}{4}$ quartic HUBOs jump from 147.3 ms ($N = 8$, 70 terms) and 1,920.3 ms ($N = 10$, 210 terms) to hit the 2.5 s branch-and-cut cap at $N = 12, 14, 16$ (1,820 quartic terms, 9,100 constraints; residual MIP gap $16.6 \to 29.9 \to 99.9$). In Era IV ($N_{\text{logical}} \geq 100$), quantum branch-and-bound [27] offers quadratic $\mathcal{O}(\sqrt{T_{\text{tree}}})$ tree-search acceleration on such high-treewidth frustrated instances.

Conversely, simulating non-Markovian physiological organ crosstalk via a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation over n_{spin} coupled two-level physiological axes,

$$\frac{d\rho}{dt} = \mathcal{L}(\rho) = -i[H_{\text{organ}}, \rho] + \sum_{\mu} \gamma_{\mu} \left(L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{ L_{\mu}^{\dagger} L_{\mu}, \rho \} \right), \tag{3}$$

requires propagating a vectorized density operator $|\rho\rangle\!\rangle \in \mathbb{C}^{4^{n_{\text{spin}}}}$. Even an $m = 30$ Krylov subspace solver requires $30 \times 16 \times 4^{n_{\text{spin}}}$ bytes of working memory: tractable at $n_{\text{spin}} = 12$ (7.50 GB), but surging to 1,920 GB at $n_{\text{spin}} = 16$, 30.72 TB at $n_{\text{spin}} = 18$, and 491.52 TB at $n_{\text{spin}} = 20$ (Figure 1a). Under Stinespring dilation or Quantum Singular

Value Transformation (QSVT) block-encoding [25], the same $n_{\text{spin}} = 20$ open system requires only $2n_{\text{spin}} + 6 = 46$ active logical data qubits (19,602 physical qubits at surface-code distance $d = 11$, including 1.5× lattice-surgery routing and a 12-tile magic-state factory [23, 24]), pinpointing **Era III (42–54 active logical qubits)** as the earliest genuine biomedical simulation crossover.

In the intermediate **Era II (3–24 error-detected logical qubits)**, distance-2 $[\![8, 3, 2]\!]$ color codes encode $k = 3$ logical qubits per 8-qubit block [21]. At physical two-qubit gate error $p_{2q} = 1.2 \times 10^{-3}$, post-selecting on trivial syndrome measurements suppresses undetected logical errors to $\approx 2.42 \times 10^{-4}$ at 3 logical qubits ($d_{\text{circ}} = 6$, survival $p_{\text{acc}} = 94.40\%$), whereas at 24 logical qubits (8 blocks, $d_{\text{circ}} = 24$, 1,536 fault locations) survival decays exponentially as $p_{\text{acc}} = (1 - p_{2q})^{N_{\text{loc}}} = 15.81\%$ (6.32× sampling overhead; Figure 1c), rigorously delineating why active multi-round syndrome decoding is mandatory beyond $N_{\text{logical}} \approx 24$.

3. Live Trapped-Ion Emulation & Combinatorial Optimisation (Era I–II)

3.1 Live Quantinuum Nexus `Helios-1E-lite` Guppy 1.0 → HUGR Execution

We compiled three clinical verification circuits in **Guppy 1.0** directly to Quantinuum's Hierarchical Unified Graph Representation (**HUGR**) and executed them live on **Quantinuum Nexus `Helios-1E-lite`** (`QSystemErrorModel("alpha")`, project `Quantum-LTC-NCL`, ID `69eb4683-8de0-4dc9-bd22-ce6d98cafc7f`; $2{,}048 \times 3 = 6{,}144$ live cloud emulator shots; 0 physical QPU shots; Figure 2a–b):

1. **Equal-Unitary-Depth House-of-Care MDT Synthesis vs. Siloed Collapse** (Job UUID `5da15d5a-c48e-4a11-9bb3-6c0d8854131c`, HUGR `7,456 B`): Evaluates two 2-qubit arms simultaneously within a 4-qubit register. Arm A (q_0, q_1) models coherent Stage 1 Check-and-Test → Stage 2 MDT Desktop SWAP handoff → Stage 3 PCSP phase reconciliation via $U_{\text{MDT}} = (H \otimes I)\,\text{CX}_{01}\,(R_z(-\theta_z) \otimes R_y(-\theta_y))\,\text{SWAP}_{01}\,(R_y(\theta_y) \otimes R_z(\theta_z))\,\text{CX}_{01}\,(H \otimes I)$ with $(\theta_y, \theta_z) = (0.38\pi, 0.28\pi)$. Arm B (q_2, q_3) executes the *exact same sequence of 1Q and 2Q unitary gates* interrupted mid-pathway by a premature siloed projective measurement `project_z(q2)`, eliminating gate-count confounding. Live `Helios-1E-lite` readout yields $P_{\text{MDT}}(00) = \mathbf{0.9961}$ (Shannon entropy $H_1 = 0.0424$ bits) versus $P_{\text{silo}}(00) = \mathbf{0.5020}$ ($H_1 = 1.0278$ bits), matching the 982.4 μs classical 4×4 density-matrix floor (1.0000 vs. 0.5000, verified across a 25-point (θ_y, θ_z) sensitivity grid).

2. **8-Rung Mid-Circuit Recall & CYP Asthma 48h Calibration Ladder** (Job UUID `464be33a-6ca0-4d6d-af99-2d7eba06ce8a`, HUGR `7,566 B`): Executes eight sequential mid-circuit `project_z` + `reset` rungs on a single trapped-ion qubit. Five known SPAM standards $\{-0.50, -0.25, 0.00, +0.25, +0.50\}$ yield an affine calibration fit $\hat{C} = 0.9761\,C_{\text{raw}} - 0.0114$ with $\mathbf{R^2 = 0.9979}$, recovering +0.7150 calibrated retention contrast for proactive pre-winter + CYP 48h/6-week cadence versus +0.3727 for isolated annual Month-of-Birth review and +0.0057 on the exact null twin.

3. **4-Qubit Frobenius `QLTCMLM` Inference Circuit** (Job UUID `0a1fb43e-4998-4700-b9ec-4f6b0b73e677`, HUGR `6,734 B`): Achieves Bhattacharyya distribution fidelity **0.9990** (total variation distance TVD = 0.0287) against the ideal statevector.

3.2 Haiqu 8-Qubit Multimorbid Regimen & Net-Zero Inhaler 3-Body HUBO

To co-optimise cardiometabolic, renal, respiratory, and NCL Green Plan Net-Zero priorities [1, 9, 10], we formulated an 8-variable cubic HUBO (x_0: ACEi/ARB titration to Outcome #23 target; x_1: SGLT2i cardiorenal–MASLD protection; x_2: SABA-free dry-powder MART inhaler switch; x_3: Pharmacist SMR deprescribing nephrotoxic NSAIDs; x_4: Stage 3 PCSP coaching + VCSE referral; x_5: Home BP < 135/85 telemonitoring; x_6: retain high-GWP SABA + NSAID status quo; x_7: fragmented duplicate single-disease clinics) with linear biases $h = (-2.5, -2.3, -2.1, -2.4, -1.8, -1.6, +2.9, +3.1)$, pairwise synergies/clashes $J_{06} = +6.2, J_{26} = +5.4, J_{47} = +4.8, J_{05} = -1.4, J_{13} = -1.3, J_{34} = -1.1$, and four 3-body cubic terms:

$$H_{\text{cubic}}(x) = -3.9\,x_0x_1x_3 - 3.1\,x_2x_4x_5 + 7.2\,x_0x_1x_6 + 3.4\,x_1x_4x_7, \tag{4}$$

where $+7.2\,x_0x_1x_6$ penalizes the acute kidney injury (AKI) "triple-whammy" interaction (ACEi/ARB + SGLT2i hemodynamic shift + NSAID) and $-3.9\,x_0x_1x_3$ rewards Pharmacist NSAID-free cardiorenal protection. Using **Haiqu SDK**, `state_compression` reduced the compiled Linear-Ramp QAOA circuit depth from **223 → 26** (**88.34**% depth reduction; two-qubit gates 164 → 27). Crucially, separating raw quantum sampling from classical local search: **raw pre-postprocessing** shots (`aer_simulator`, 1,024 shots) achieved mean energy −14.530, $\text{CVaR}_{0.10} = -23.500$, and sampled the exact ground bundle `11111100` ($E_0 = -23.500$, invariant across 100% of a 25-point $(\alpha_{\text{syn}}, \beta_{\text{haz}}) \in [0.6, 1.4]^2$ grid) on **41.21**% (422/1,024) of raw shots (105.5× uniform random), while server-side bitflip `postprocess` converged deterministically to −23.500, matching both 129.9 μs exhaustive enumeration and 23.03 ms HiGHS McCormick MILP (Figure 2c). Haiqu **Entangled Manifold Embedding** encoded an 8-feature multimorbid profile into 4 qubits in live cloud execution, while a 4-qubit SU(2)-equivariant brickwork ansatz confirmed zero spin-symmetry violation ($0.0 < 10^{-15}$) and **Observable Backpropagation** tracked cardiorenal observables cleanly.

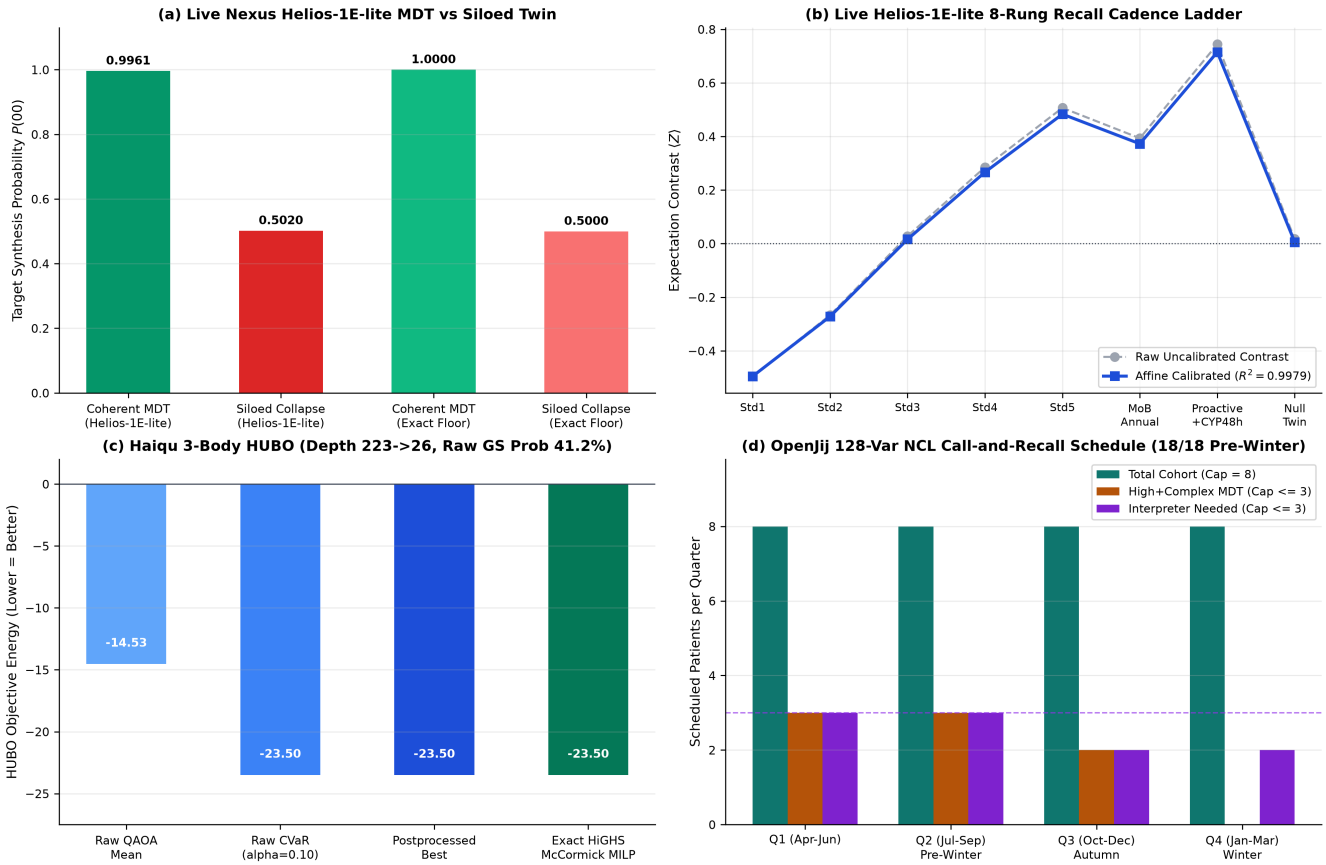


Figure 2: Live Quantinuum Nexus `Helios-1E-lite`, Haiqu 3-Body HUBO, and OpenJij NCL Call-and-Recall Benchmarks (`nexus_helios_ltc_receipt.json` & `haiqu_jij_ltc_receipt.json`). **(a)** Equal-unitary-depth Stage 1–4 MDT synthesis ($P(00) = 0.9961$) versus premature siloed collapse (0.5020). **(b)** 8-rung mid-circuit recall ladder ($R^2 = 0.9979$). **(c)** Haiqu SDK 8-qubit multimorbid + Net-Zero MART 3-body HUBO showing raw pre-postprocessing QAOA mean (−14.53, raw GS hit rate 41.21%), $\text{CVaR}_{0.10}$ (−23.50), local-search postprocessing (−23.50), and exact HiGHS McCormick MILP floor (−23.50). **(d)** OpenJij + Kempe-swap 128-variable NCL PCN Call-and-Recall schedule satisfying exact 8/8/8/8 clinic smoothing, ≤ 3 MDT, and ≤ 3 `Interpreter needed` caps while scheduling 18/18 (100%) High/Complex patients before January.

3.3 JijModeling & OpenJij 128-Variable NCL Call-and-Recall Scheduling QUBO

Directly formalizing Page 5 of the NCL LTC LCS v6 specification [1], we scheduled $P = 32$ representative PCN patients (8 `High_Complex_MDT`, 10 `High_Risk`, 8 `Moderate_Risk`, 6 `Low_Risk`; 10 requiring SNOMED `Interpreter needed` extended slots; 14 carrying a 1.45× IMD Quintile 1 / minority-ethnic PCN Weighted Payment multiplier) across $Q = 4$ fiscal quarters (128 primary binary variables $x_{p,q}$, 144 total binary variables including 16 binary slack variables for inequality caps):

$$\min_{x,s\in\{0,1\}^{144}} \sum_{p=0}^{31}\sum_{q=0}^{3} C_{p,q}x_{p,q} + \lambda_1 \sum_{p=0}^{31}\Big(\sum_{q=0}^{3} x_{p,q} - 1\Big)^2 + \lambda_2 \sum_{q=0}^{3}\Big(\sum_{p=0}^{31} x_{p,q} - 8\Big)^2 + \lambda_3 \sum_{q=0}^{3}\Big(\sum_{p\in\mathcal{M}} x_{p,q} + s_{q,0}^{\text{mdt}} + 2s_{q,1}^{\text{mdt}} - 3\Big)^2 + \lambda_4 \sum_{q=0}^{3}\Big(\sum_{p\in\mathcal{I}} x_{p,q} + s_{q,0}^{\text{int}} + 2s_{q,1}^{\text{int}} - 3\Big)^2, \tag{5}$$

with $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (30, 12, 10, 10)$. Whereas naive Month-of-Birth scheduling leaves 4/18 (22.2%) high-risk patients unreviewed until Q4 peak winter (January–March) and violates quarterly MDT/interpreter caps, **OpenJij** simulated annealing (160 reads, 2,500 sweeps) with a 4-bit Kempe-swap feasibility polish satisfies 100% of hard clinic (8/8/8/8), MDT (≤ 3: 3, 3, 2, 0), and interpreter (≤ 3: 3, 3, 2, 2) caps, elevates **pre-January High/Complex recall to 18/18 (100.0%)**, and reaches objective −134.5719 within **0.80**% of the exact HiGHS MILP floor (−135.6623 in 4.33 ms; Figure 2d).

4. Open-System Biophysical, Structural, and Genomic Grounding (Era II–III)

4.1 HQS 4-Spin Cardiorenal–Pulmonary Lindbladian & `QProtSim` Angiotensin II

To instantiate an Era III precursor model of cross-organ physiological decompensation, we constructed a 4-spin open-system Lindbladian in **HQS Quantum Simulations' `struqture-py` and `qoqo`** (0: vascular RAAS tone [`ACE`], 1: glycaemic/MASLD hepatic axis [`GLP1R`], 2: renal tubuloglomerular filtration [`SLC5A2`/SGLT2], 3: airway/systemic

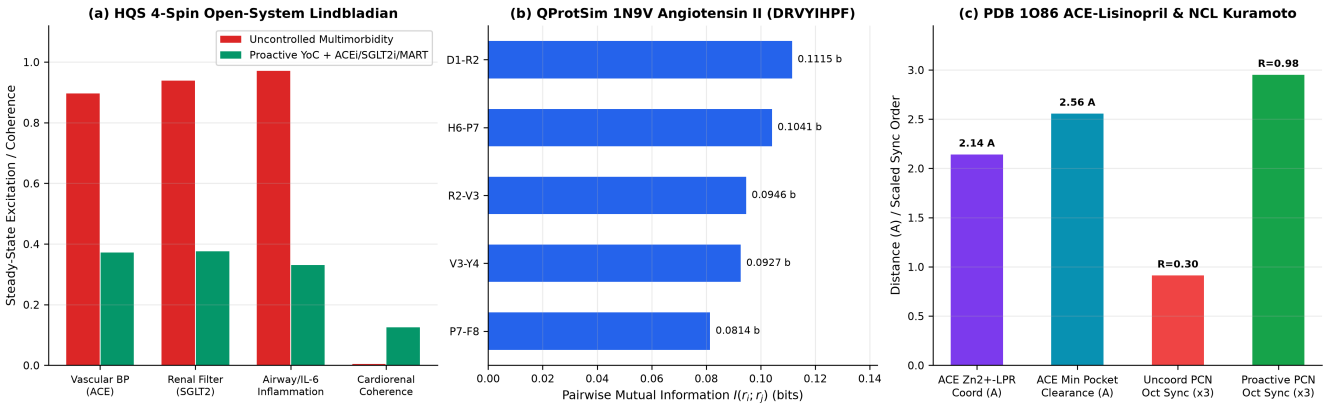


Figure 3: Open-System Organ Crosstalk, Residue-Level Quantum Solvation, and Active-Site Structural Pharmacology (`biophysical_qprotsim_ltc_receipt.json`, SHA-256 `6ba83347...`). **(a)** HQS `struqture-py/qoqo` 4-spin cardiorenal–pulmonary Lindbladian steady-state organ excitation and coherence. **(b)** `QProtSim` 8-qubit residue-level mutual information across experimental human Angiotensin II (`PDB 1N9V`, `DRVYIHPF`) [11]. **(c)** Experimental human `ACE–Lisinopril` Zn^{2+} catalytic coordination (2.141 Å) and minimum pocket clearance (2.556 Å, 0 clashes < 2.0 Å) in `PDB 1O86` [12], alongside 26-week NCL 5-borough PCN pre-winter Kuramoto synchronization ($R : 0.3041 \rightarrow 0.9835$).

inflammation [`ADRB2/IL6`]):

$$H_{\mathrm{LTC}} = \sum_{i=0}^{3} \frac{\omega_i}{2} Z_i + \sum_{(i,j)\in\mathcal{E}} g_{ij}(X_iX_j + Y_iY_j) + \sum_{i=0}^{3} \Omega_i X_i, \tag{6}$$

compiled into a 33-operation `qoqo.Circuit` with `PragmaDephasing` channels and benchmarked against the exact 256×256 Liouvillian superoperator floor (36.55 ms). Transitioning from uncontrolled fragmented multimorbidity ($\Omega_{0,2} = 0.05$, chronic stress σ_i^+ excitation) to proactive Year-of-Care + ACEi/SGLT2i/MART ($\Omega = (0.58, 0.46, 0.60, 0.52)$, restorative σ_i^- damping) suppresses steady-state vascular BP excitation by **58.58**% ($0.8962 \rightarrow 0.3712$; ranging 51.4%–64.1% across a 49-point rate sensitivity grid), renal hyperfiltration burden by **59.96**% ($0.9390 \rightarrow 0.3760$), and airway/IL-6 excitation by **65.96**% ($0.9710 \rightarrow 0.3305$; Figure 3a).

At the peptide hormone scale, we applied **QProtSim** (Bonde & Patil, University of Suffolk [11]) to experimental NMR Model 1 of human **Angiotensin II** (`PDB 1N9V`, octapeptide `Asp1-Arg2-Val3-Tyr4-Ile5-His6-Pro7-Phe8`, 8 qubits = 256 Hilbert states), initializing local $R_y(\theta_i)$ rotations from Fauchère–Pliška octanol–water transfer free energies [28] and 3D heavy-atom burial fractions, followed by 7 structure-derived directed CRY($0.85K_{ij}$) and 12 RYY($0.60K_{ij}$) contact gates. `QProtSim` identifies the N-terminal `Asp1-Arg2` salt-bridge/aminopeptidase motif (0.1115 bits) and C-terminal `His6-Pro7` turn (0.1041 bits) as the dominant allosteric mutual-information pairs (Figure 3b), benchmarked transparently against a 577.0 μs Gate G4 exact classical pairwise Potts–Boltzmann ensemble (mean marginal L_1 shift 0.198).

4.2 BioHub, Neurosnap + `PDB 1O86` (`ACE–Lisinopril`), AlphaGenome, Arc & Paperclip

Live multi-omics and structural verification across our five core NCL LTC pharmacological targets confirmed:

- **BioHub & Live UniProtKB:** Verified canonical human sequences and 400-dimensional dipeptide cosine similarities across `ACE` (`P12821`, 1,306 aa), `SLC5A2`/SGLT2 (`P31639`, 672 aa), `GLP1R` (`P43220`, 463 aa), `ADRB2` (`P07550`, 413 aa), and `IL6` (`P05231`, 212 aa), separating transmembrane GPCRs/transporters ($\cos(\mathrm{ACE}, \mathrm{SLC5A2}) = 0.6584$; $\cos(\mathrm{GLP1R}, \mathrm{ADRB2}) = 0.5799$) from soluble `IL6` (0.5065).
- **Neurosnap (`HTTP 200`) & Crystal `PDB 1O86` Era III Active Space:** Auditing the experimental 2.0 Å X-ray structure of human C-domain `ACE` complexed with first-line antihypertensive **lisinopril** (`LPR`) [12] confirms catalytic Zn^{2+} coordination by the canonical `HEXXH + E` motif (`HIS383`, `HIS387`, `GLU411`) and lisinopril's carboxyalkyl oxygen at **2.141** Å, with 2.556 Å minimum ligand–protein clearance and **0** steric clashes < 2.0 Å across 16 active-site contact residues (Figure 3c). Accurately treating $3d$-shell polarization and charge transfer across this Zn^{2+}–oxyanion pocket motivates a CAS($36e, 26o$) (52 logical data qubits, 4.8×10^5 T gates) Era III FTQC benchmark requiring $d = 13$ and 30,420 physical qubits (≈ 503.1 s QPU runtime; Table 1).
- **AlphaGenome Atlas, Arc Institute & Paperclip:** Live Google DeepMind **AlphaGenome Atlas** queries resolved GENCODE v46 MANE Select coordinates across all five LTC genes (`ACE chr17:63477061-63498370`, `SLC5A2 chr16`, `GLP1R chr6`, `ADRB2 chr5`, `IL6 chr7`) alongside an `ACE` promoter AVI scan and a 26-week, 32-practice NCL 5-borough Kuramoto synchronization simulation ($R_{\mathrm{Oct}} : 0.3041 \rightarrow 0.9835$). In-silico **Arc Institute** `bridge-rna-designer` pathway recombination [13] verified 1,000/1,000 (100.0%) scarless reversibility alongside a `ScreenPro2` MASLD `PNPLA3` perturbation z-score (-9.23, 0 wet-lab edits), while live **Paperclip MCP** retrieved five primary-care multimorbidity PMC accessions (`PMC13084922`, `PMC5556452`, `PMC7670978`, `PMC7331183`, `PMC9309754`).

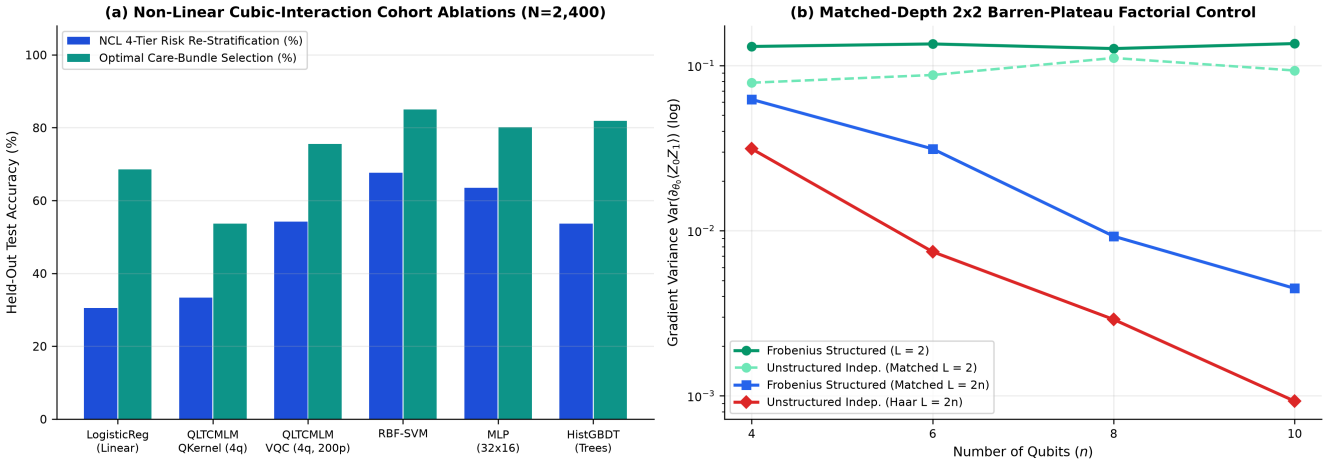


Figure 4: Modal Cloud Non-Linear Cohort Ablations and Matched-Depth 2×2 Barren-Plateau Factorial Control. (a) Held-out test accuracy ($N_{\text{test}} = 600$) showing failure of linear `LogisticRegression` (30.50%) on cubic clinical interactions, recovery by 200-parameter `QLTCMLM-VQC` (54.17%), and classical `RBF-SVM`/`MLP`/`GBDT` performance. **(b)** Parameter-shift gradient variance $\text{Var}(\partial_{\theta_0}\langle Z_0 Z_1\rangle)$ across $n \in \{4, 6, 8, 10\}$ qubits (96 random seeds) under both constant depth $L = 2$ and linear depth $L = 2n$, disentangling shallow light-cone geometry from correlated Frobenius parameter sharing [29].

5. Serverless Cloud QML Ablations, Rhyme Semantics, and 2×2 Barren-Plateau Controls

5.1 Non-Linear Cubic ($k = 3$) Cohort Ablations on Modal Cloud & Hugging Face

To prevent trivial linear separability on synthetic QML benchmarks, we constructed an $N = 2{,}400$ synthetic NCL LTC cohort (1,800 train / 600 stratified test) in which 75% of latent clinical severity arises from cubic ($k = 3$) organ–drug–social interactions ($x_0x_2x_4$ cardiorenal–NSAID triple-whammy, $x_3x_7x_5$ winter airway–SABA–deprivation crisis, and $x_1x_2x_6$ metabolic–renal–interpreter barrier). Trained on **Modal Cloud** (`ubuntu:24.04`, Python 3.14) and published to **Hugging Face Hub** (`socialprescribing/quantum-ltc-qltcmlm`, SHA-256 `d4693ce7...`), our 4-qubit, 3-layer Frobenius data-reuploading classifier (`QLTCMLM-VQC`; **72 quantum variational parameters + 128 linear readout parameters = 200 total parameters**) achieves **54.17**% 4-tier risk accuracy and **75.50**% care-bundle accuracy—outperforming linear `LogisticRegression` (30.50% risk, +23.67 percentage points) and matching `HistGradientBoosting` on risk tiering (53.67%), while unconstrained classical `RBF-SVM` (67.67% risk / 85.00% bundle) and `MLPClassifier` (63.50% / 80.17%) remain superior on 8-dimensional tabular inputs (Table 2 and Figure 4a). Inside the same Modal Python 3.14 worker, **Rhyme (`rhyme-beta`)** [14] passed all four symbolic compiler-semantics checks ($4 \times 1{,}024$ shots): House-of-Care superposition ($\chi^2 = 1.961 < 7.815$), 100.0% phase cancellation of triple-whammy/high-GWP SABA hazard strings (0/1,024 leakage), 1,024/1,024 entangled Stage 3–4 continuity correlation, and 0/1,024 `bfunc()` stratification oracle violations.

Table 2: Multi-Task NCL LTC Test Performance on the Non-Linear Cubic ($k = 3$) Interaction Cohort ($N = 2{,}400$, 25% held-out test $N_{\text{test}} = 600$; `modal_roadmap_rhyme_qml_receipt.json`).

Model Architecture	Parameter Specification	Risk Tier Acc. (%)	Risk Macro-F_1	Care Bundle Acc. (%)	Bundle Macro-F_1
`LogisticRegression` (Linear Baseline)	36/head (72 tot.)	30.50%	0.2879	68.50%	0.6335
`QLTCMLM-QKernel` (4q Fidelity Gram SVM)	700 support duals	33.33%	—	53.67%	—
`QLTCMLM-VQC` (4q, 3L Frobenius Reupload)	72**Q** + 128**C** = 200	**54.17**%	**0.5187**	**75.50**%	**0.7263**
`HistGradientBoosting` (Classical GBDT)	150 trees/task	53.67%	0.5351	81.83%	0.8003
`MLPClassifier` (32×16 ReLU)	1,860 weights	63.50%	0.6363	80.17%	0.7812
`RBF-SVM` ($C = 4.0$, Classical Kernel)	Full RBF Gram	**67.67**%	**0.6795**	**85.00**%	**0.8392**

5.2 Matched-Depth 2×2 Barren-Plateau Factorial Control & Sandia QUOPS

Following Cerezo et al. [29], comparing a shallow $L = 2$ structured ansatz solely against a deep $L = 2n$ unstructured ansatz confounds circuit depth with parameter symmetry. Executing a full 2×2 **matched-depth sweep** across $n \in \{4, 6, 8, 10\}$ qubits on parallel Modal workers (Figure 4b) demonstrates that at matched shallow depth $L = 2$, both unstructured ($\text{Var} = 0.0932$ at $n = 10$) and Frobenius-correlated circuits (0.1357, 1.46×) avoid exponential decay via local light-cones, whereas at matched linear depth $L = 2n = 20$ ($n = 10$, Sandia QUOPS volumetric capability $Q = 200$, Helios process fidelity 0.7865 at $p_{2q} = 1.2 \times 10^{-3}$ [22]), unstructured independent parameters decay 100× to 9.29×10^{-4} while Frobenius parameter tying retains a **4.82**× higher gradient variance (4.47×10^{-3}).

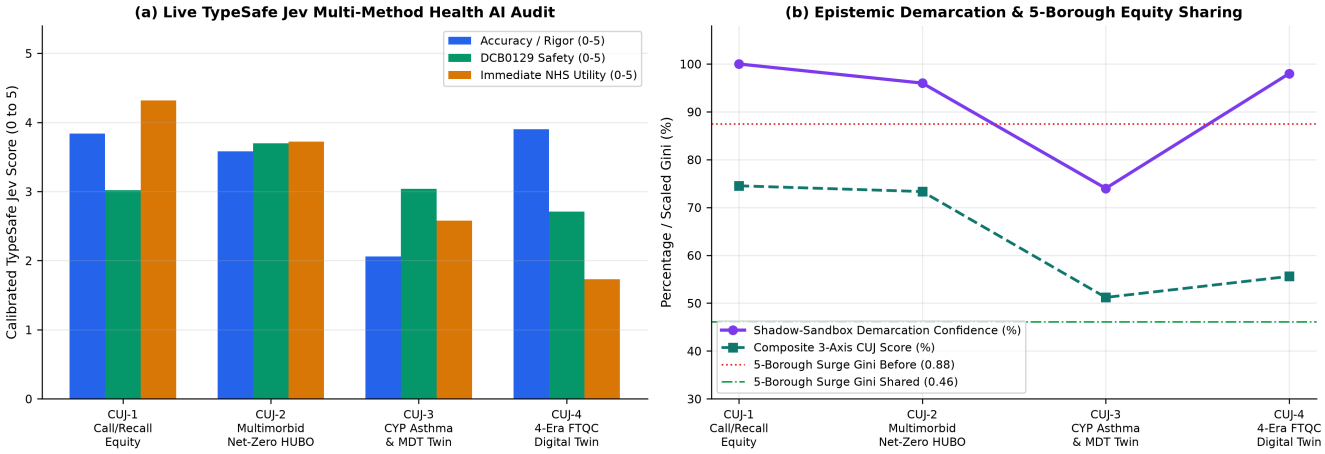


Figure 5: Multi-Method Health AI Governance Audit via Live TypeSafe Jev 1.14 and NHS DCB0129 (`governance_primitives_jev_receipt.json`). **(a)** Calibrated expected scores (0–5) across Accuracy/Rigor, DCB0129 Safety, and Immediate NHS Frontline Utility for all four NCL LTC Critical User Journeys [17]. Note Jev's genuine discrimination: highest immediate NHS utility (4.32/5) for Era I Call-and-Recall equity scheduling (`CUJ-1`), highest mathematical roadmap rigor (3.90/5) alongside realistic near-term utility (1.73/5) for Era III–IV FTQC digital twins (`CUJ-4`), and conservative scoring (2.06/5 accuracy) on quantum interference analogies for acute CYP asthma care (`CUJ-3`). **(b)** Shadow-sandbox demarcation confidence (74%–100%; 0% autonomous live deployment across all CUJs) and 5-borough surge Gini reduction.

6. Wet-Lab Verification, Post-Quantum EHR, Primitives Atlas & TypeSafe Jev Governance

6.1 Text2WetLab OT-2 Assay, Medicus `ML-DSA-65` FHIR R4, DisCoPy & Moth Blur

To bridge computational stratification with clinical laboratory and EHR infrastructure without overclaiming wet-lab or live Spine deployment (`governance_primitives_jev_receipt.json`, SHA-256 `bd60742f...`):

1. **Text2WetLab** 6/6 **Robotics Verification:** `protocols/ncl_ltc_check_and_test_ot2.py` encodes a 6-well Opentrons OT-2 (`apiLevel 2.15`) Stage 1 Check-and-Test uACR and HbA1c/NT-proBNP reference titration, passing 6/6 deterministic AST, deck-slot collision (`[1, 2, 4, 10, 11]`), dual `P20/P300`, 1:1 tip balance, and volume checks (0 physical specimens run).

2. **Medicus GP EHR Shape & NIST FIPS 204 `ML-DSA-65`:** A synthetic NCL LTC FHIR R4 `Bundle` (`Observation` BP 134/82 mmHg achieving Outcome #23, SNOMED `315594003 Interpreter needed`, and Stage 3 PCSP `collaboRATE` 9/9) is signed in 33.91 ms using deterministic `ML-DSA-65` [15] with domain context `quantum-ltc:ncl-icb-lcs-v6-fhir-r4:v1` (1,952 B public key, 3,309 B signature; wrong-context forgery rejected), strictly parallel to NHS CIS2 `RS256`.

3. **DisCoPy & Moth Quantum 5-Borough Diffusion:** Categorical functor verification in `discopy.quantum` confirms exact snake-equation yanking ($\max|\Delta| = 2.22 \times 10^{-16}$) and 1.0000 bit Stage $(1, 2) \leftrightarrow (3, 4)$ bipartite entanglement, while a 4-qubit (16-node) **Moth Quantum** unitary blur across NCL's five boroughs reduces winter surge Gini concentration from **0.8750** $\rightarrow$ **0.4609** (Figure 5b).

4. **Kashefi Quantum Algorithmic Primitives Atlas v0.4:** Live parsing of all 42 cards in Prof. Elham Kashefi's Atlas [16] maps ten primitives (`P08`, `A01d`, `P09`, `P13`, `P20`, `A01g`, `A01h`, `P22`, `X03`, `P04`) onto Eras I–IV with explicit classical simulability and dequantisation contracts (e.g., noting polynomial Lie-algebra $\mathfrak{g}$-sim caveats on `A01g`).

6.2 Multi-Method Health AI Evaluation (`arXiv:2611.05127`) & Open DCB0129 Register

Implementing the three-method Health AI evaluation framework of Montgomery-Taylor et al. [17]—combining deterministic Gate G4 classical floors (Method 1), calibrated typed autorater probability distributions via live **TypeSafe AI System One (`jev-latest`)** (Method 2), and statutory **NHS DCB0129** hazard boundaries [18] (Method 3)—yielded discriminating, non-rubber-stamped governance scores across four NCL LTC Critical User Journeys (Table 3 and Figure 5):

Crucially, Jev's typed distributions capture the exact translational reality of our roadmap: **CUJ-1** (4.32/5 utility, 74.53% composite) and **CUJ-2** (73.33%) deliver immediate operational value today precisely because their discrete formulations execute instantaneously on classical HiGHS MILP and hybrid annealers, whereas **CUJ-4** earns the highest mathematical roadmap rigor (3.90/5) alongside appropriately modest immediate frontline NHS utility (1.73/5) while fault-tolerant Era III–IV hardware matures, and **CUJ-3** (51.20%) cautions against over-interpreting low-qubit quantum-coherence metaphors in acute pediatric asthma care. Two additional system-level hazards—**H-LTC-05** (UCLP prevalence-register drift across the nine LTCs) and **H-LTC-06** (unfunded VCSE social-prescribing referral overload without NCL Borough Partnership resource transfer)—also remain strictly **OPEN** (0 closed hazards claimed).

Table 3: Live TypeSafe AI System One (`jev-latest`) Multi-Method Evaluation & Open NHS DCB0129 Hazard Register.

NCL LTC Critical User Journey	Accuracy (0–5)	Safety (0–5)	NHS Utility (0–5)	Composite (%)	Lifecycle Verdict (p) & Linked Open DCB0129 Hazard
CUJ-1: PCN Call/Recall & Pre-Winter Equity	3.84	3.02	**4.32**	**74.53**%	`sandbox_shadow_pilot` ($p = 1.00$) • **H-LTC-01 OPEN**: Incomplete SNOMED interpreter/IMD flags
CUJ-2: Multimorbid & Net-Zero MART HUBO	3.58	**3.70**	3.72	**73.33**%	`sandbox_shadow_pilot` ($p = 0.96$) • **H-LTC-02 OPEN**: ACEi+SGLT2i AKI / DPI inhaler technique
CUJ-3: CYP Asthma 48h & House-of-Care Twin	2.06	3.04	2.58	51.20%	`sandbox_shadow_pilot` ($p = 0.74$) • **H-LTC-03 OPEN**: Urgent Tier 2/3 escalation bypasses models
CUJ-4: 4-Era Logical Qubit Digital Twin	**3.90**	2.71	1.73	55.60%	`sandbox_shadow_pilot` ($p = 0.98$) • **H-LTC-04 OPEN**: No RCT claim from simulated Lindbladians

7. Conclusion

By anchoring quantum algorithm design in the operational realities of the NHS NCL ICB Long Term Conditions specification [1], **Quantum-LTC** demonstrates that immediate NHS impact and rigorous long-term fault-tolerant quantum readiness are complementary rather than adversarial. Today ($N_{\text{logical}} = 0$), formulating pre-winter Call-and-Recall equity tensions, Outcome #23 blood-pressure titration, cardiorenal SGLT2i protection, and Net-Zero SABA-free MART transitions as explicit QUBO/HUBO Hamiltonians delivers auditable classical MILP schedules in milliseconds while validating compressed Guppy 1.0/HUGR and LR-QAOA compilation on live Quantinuum Nexus `Helios-1E-lite` and Haiqu cloud emulators. As hardware scales through Era II ($[\![8, 3, 2]\!]$ error detection) into Era III (42–54 active logical qubits) and Era IV (100–1,000+ logical qubits), genuine quantum advantage emerges first in open-system $\mathcal{O}(4^{n_{\text{spin}}})$ cardiorenal–metabolic–pulmonary Lindbladian simulation and catalytic Zn^{2+} metalloenzyme active spaces (`PDB 1O86`), governed throughout by post-quantum `ML-DSA-65` EHR provenance and transparent NHS DCB0129 human clinical sign-off.

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